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Simple linear models of coastal setup and seicheing behavior across the Laurentian Great lakes

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ABSTRACT

Storm surges and seiches are two dynamically distinct behaviors of lakes that contribute to coastal sea level fluctuations on short time scales and can have significant societal and ecological impacts. Most of the emphasis in the existing Laurentian Great Lakes literature on storm surges focuses on wind-driven storm surges on Lake Erie, due to their large magnitudes and major population centers at both ends of the lake. On Lake Superior, however, the primary driver of coastal setup is atmospheric pressure gradients, which depress water levels at one end and increase them at the other. Lakes Michigan, Huron, and Ontario are intermediate, and both wind forcing and atmospheric pressure gradients contribute to coastal setup. The primary driver of this difference in response is lake depth, which is explored here using simple linear theory. In addition, lake depth plays an important role in the character of the subsequent seiche response of lakes; in shallow lakes such as Erie, bottom drag can damp out seicheing behavior quickly, whereas in deep lakes like Superior, seiches can persist for many oscillations. A survey of large lakes worldwide suggests a wide range of behaviors.

1. Introduction

Coastal setup of lake level is an issue of great societal importance; large setup events can result in inundation of populated areas, resulting in significant impacts to infrastructure (Danard et al., 2003; Gronewold et al., 2013; Meadows et al., 1997; Shepard et al., 2012), as well as reciprocal low water levels at the other end of the lake, which can, for example, expose water intakes. Coastal setup and seiches are also acknowledged to play a role in coastal wetland ecosystems (Bedford, 1992; Linares et al., 2018; Trebitz, 2006). For all of the attention merited storm surges, the majority of the work done on the Laurentian Great Lakes (hereafter “the Great Lakes”) has been done on a single lake. Lake Erie, because of its shallowness, exhibits much larger surges than the other, deeper, Great Lakes. Past work on Erie has focused on specific storm events (Dingman and Bedford, 1984; Donn, 1959; Hamblin, 1979), on empirical models relating wind stress to setup (Hamblin, 1987), on the development of numerical modeling tools to improve coastal forecasting (Schwab, 1978), and more recently on the potential for use in energy generation (Farhadzadeh et al., 2017). In Erie, the link between wind stress and setup is so strong that it has been used to

improve estimates of wind stress over the lake (Drews, 2013; Schwab, 1982).

The focus on Erie has resulted in the neglect of a second process for driving coastal water level fluctuations in the Great Lakes literature. The inverted barometer effect (Gill, 1982; Wunsch and Stammer, 1997) results in water level fluctuations due to gradients in atmospheric pressure, which can depress water on one end of the lake and raise water levels at the other end. To the extent that it is mentioned, it is largely to point out that it can safely be ignored in Lake Erie, due to the relative size of the wind forced response (Dingman and Bedford, 1984; Hamblin, 1987). However, in deeper lakes where the wind driven response is weak, the inverted barometer effect can actually dominate coastal forced water level fluctuations, as acknowledged by Kelly et al. (2024) and as is shown here. While these two mechanisms are considered, we here disregard fluctuations due to wave setup, which during large wave events can be significant (Drews, 2013; Lentz and Raubenheimer, 1999).

Contrary to the single-lake emphasis with regards to storm surges, the oscillatory seiche response following setup has been more broadly studied. Early numerical and observational work on Superior (Platzman 1972), Michigan and Superior (Mortimer et al., 1976), Ontario

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(Hamblin, 1982), and Ontario and Superior (Rao and Schwab, 1976) all focused on determining the frequencies of the primary and higher order seiche modes (this problem appears to be one of the earliest widely pursued applications of numerical modeling to the Great Lakes). Little if any work has been done examining the role that bottom drag may play in damping out the seiche response, although Simons and Schertzer (1989) acknowledge the importance of bottom drag in understanding the response of shallow Lake St. Clair. The (e-folding) time scale of seiche damping in Lake Superior was estimated by Kelly et al. (2024) to be roughly 4.5 days, roughly consistent with this work. We will consider linear drag in this case; while quadratic drag (and more sophisticated formulations) is acknowledged to be more realistic than linear drag (e.g. Shimizu and Imberger, 2008), using a linear formulation allows the derivation of a simple analytical solution that provides insight into the role of depth in the damping of the seiche. Depth plays multiple roles in the forced (setup) and free (seiches) response of a lake: In deep lakes, the wind driven response for a given applied wind stress will be weak compared to the same response in a shallow lake; the period of the seiche is a function of the depth; and bottom drag plays a larger role in shallow lakes, meaning that while setup may be smaller in magnitude in deep lakes, the resulting seiche is likely to last longer than in a shallow lake.

In this short note, we develop a basic dynamical understanding of these relationships by using an idealized application of the one-dimensional, non-rotational shallow water equations, which assume that lateral velocities are uniform with depth. Obviously, the great lakes are not simple, one-dimensional basins; however, we can use these concepts to develop a better dynamical understanding of how free and forced displacement of coastal water levels functions across the lakes. As will be shown, even these very simple analytical models are capable of explaining the response of these lakes remarkably well. While numerical models can be applied to provide a much more quantitatively accurate representation of setup events and the resulting seiche response, the very simple dynamical approach taken here can be used to build intuition about how the lakes respond to forcing. Ultimately, the purpose of this short contribution is not to quantitatively improve upon previous works, which in general take much more sophisticated approaches to the problem, but to provide a simple dynamical framework that can improve understanding. This paper will consider all five lakes, and Superior and Erie will be emphasized as examples of a deep and shallow lake, respectively. While the basis of much of the work presented here can be found in various engineering textbooks (e.g. Kundu et al., 2012), note that most limnology texts (e.g. Hamilton and MacIntyre, 2024) do not cover the math presented here. The results here are also scientifically novel in that we systematically consider the role of the inverted barometer term, as well as the role of damping in the behavior of seiches across the Great Lakes. If nothing else, this paper should be a useful teaching tool when addressing basic barotropic dynamics in the Laurentian Great Lakes.

2. Methods

2.1. Theory

The non-rotational, one-dimensional, shallow water momentum equation is:

$$\frac{du}{dt} = -g \frac{d\eta}{dx} - \frac{1}{\rho} \frac{dp_{ATM}}{dx} - \frac{r}{H} u + \frac{\tau^s}{\rho H} \quad (1)$$

(Cushman-Roisin and Beckers, 2011), where $u(x,t)$ the vertically uniform velocity along the channel, t is time, g the acceleration due to gravity, $\eta(x,t)$ the surface displacement from equilibrium, x the lateral coordinate system of the basin where the two ends of the lake of length L are characterized by $x = 0$ and $x = L$, $\rho = 1000 \text{ kg m}^{-3}$ is the density of water, p_{ATM} the atmospheric pressure, H the uniform water depth, r a linear drag constant, taken here to be $5 \times 10^{-4} \text{ m s}^{-1}$ (Lentz and Winant,

1986), and τ^s is surface wind stress along the axis of the lake. The terms in equation (1), in order, represent acceleration, the pressure gradient due to surface displacement, the atmospheric pressure gradient, bottom drag, and surface wind stress. In addition, the continuity equation can be written as:

$$\frac{d\eta}{dt} + H \frac{du}{dx} = 0 \quad (2)$$

Different dynamics can be explored by balancing various terms of the momentum equation (1) against each other.

We start by considering the forced response of a lake to wind and atmospheric pressure gradients. In the case of wind-driven setup, the primary equilibrium balance is between the surface stress and the pressure gradient term. In this case, the balance becomes

$$0 = -g \frac{d\eta}{dx} + \frac{\tau^s}{\rho H}$$

So that the resulting surface slope is

$$\frac{d\eta}{dx} = \frac{\tau^s}{\rho g H}$$

In a basin of length L , the water level difference across the basin will therefore be

$$\Delta\eta_{wind} = \frac{L\tau^s}{\rho g H} \quad (3)$$

The observed setup/setdown at either end of the lake will be half of this. While this appears straightforward, it is complicated by the fact that the stress over lakes as large as these is not uniform; meteorological stations measure wind speed, not wind stress; and that the relationship between speed and stress is a complex function of the air–water temperature difference (Schwab, 1978). However, the basic functional dependencies are clear: large wind events will result in large setup, and deeper lakes will experience less setup than shallow lakes given the same surface stress.

The impact of the inverted barometer (IB) effect can be determined by assuming that at equilibrium the horizontal pressure gradient internal to the lake is zero (Gill, 1982); the atmospheric pressure gradient and the pressure gradient due to the sloping surface cancel each other out. In this case,

$$\rho g \frac{d\eta}{dx} = - \frac{dp_{ATM}}{dx}$$

so that the resulting water level difference is

$$\Delta\eta_{IB} = - \frac{1}{\rho g} \Delta p_{ATM} \quad (4)$$

Where Δp_{ATM} is the atmospheric pressure difference across the lake. Assuming that the magnitude of atmospheric pressure gradients is similar across the great lakes, this pressure difference is going to be larger for larger lakes, so we might expect the inverted barometer effect to be more important on larger (longer) lakes. Again, the observed setup at one end of the lake due to this effect will be half of this value. The two forcing mechanisms are uncorrelated with each other, although the pressure difference is correlated with the transverse component of the wind, which is to be expected given the geostrophic nature of large-scale atmospheric flows.

Next, we consider the free, unforced response, which often follows a setup event. The primary balance becomes

$$\frac{du}{dt} = -g \frac{d\eta}{dx} - \frac{r}{H} u$$

When combined with the continuity equation by cross-differentiation

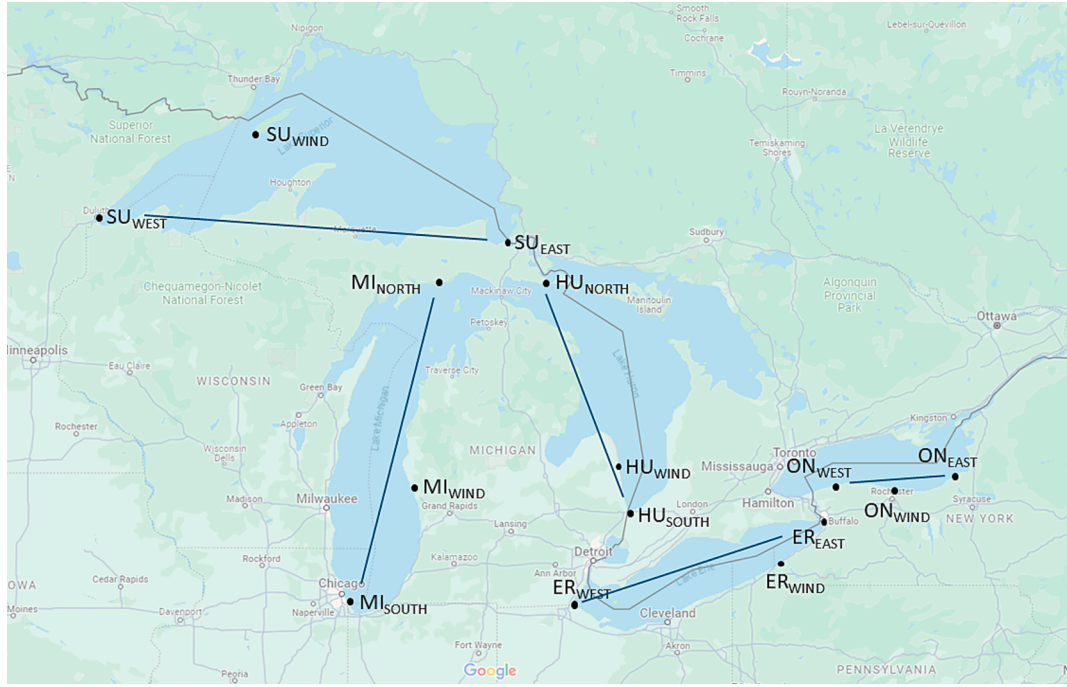


Fig. 1. Location of coastal meteorology stations used for barometric pressure and wind velocity and water level stations.

this becomes

$$\frac{d^2\eta}{dt^2} = gH \frac{d^2\eta}{dx^2} - \frac{r}{H} \frac{d\eta}{dt}$$

which is the classical wave equation with an additional damping term; it has decaying wavelike solutions. When applied to a lightly damped closed basin of length L (which is the case for all five Great Lakes), the solution for surface displacement is the superposition of standing waves, the lowest mode of which is

$$\eta(x, t) = A \exp(-t/t_{DEC}) \cos\left(\frac{2\pi}{t_{OSC}} t\right) \cos\left(\frac{\pi}{L} x\right)$$

Where A is an amplitude (determined by the precedent forcing), t_{DEC} is a decay time scale (e-folding scale of the oscillation amplitude), and t_{OSC} the seiche period.

At $x = 0$ this becomes

$$\eta(0, t) = A e(-t/t_{DEC}) \cos\left(\frac{2\pi}{t_{OSC}} t\right)$$

which represents oscillatory behavior with an exponentially decaying amplitude. The solution at the other end of the lake ($x = L$) is

$$\eta(L, t) = -A e(-t/t_{DEC}) \cos\left(\frac{2\pi}{t_{OSC}} t\right) = -\eta(0, t)$$

So that the solution at the two ends of the lake are 180° out of phase.

For the lowest (primary) mode, the period t_{OSC} is

$$t_{OSC} = \frac{2L}{\sqrt{gH - \left(\frac{rL}{2\pi H}\right)^2}}$$

Which, for weak bottom drag, reduces to the more familiar

$$t_{OSC} = \frac{2L}{\sqrt{gH}} \quad (5)$$

often referred to as Merian's Formula (e.g. Hamilton and MacIntyre,

Table 1

Geographic parameters and data sources for five lakes.

Lake	L (km)	H (m)	Station names	Barometric pressure	Water Level	Wind
Superior	570	147	SU_WEST	DULM5	9099064	ROAM4
			SU_EAST	PTIM4	9099004	
			SU_WIND			
Michigan	490	85	MI_SOUTH	CMTI2	9087044	MKGM4
			MI_NORTH	PNLM4	9087096	
			MI_WIND			
Huron	350	59	HU_NORTH	DTLM4	9075099	HRBM4
			HU_SOUTH	FTGM4	9014098	
			HU_WIND			
Erie	400	19	ER_WEST	THRO1	9063085	DBLN6
			ER_EAST	BUFN6	9063020	
			ER_WIND			
Ontario	180*	86	ON_WEST	OLCN6	9052076	RPRN6
			ON_EAST	OSGN6	9052030	
			ON_WIND			

* On Lake Ontario, there are no water level stations at the ends of the lake (Fig. 1). When evaluating eqn. (5) for the seiche period, a length of 310 km is used.

2024), which is the period of the seiche. These two expressions are nearly identical for values of H and L for most lakes, and we will use eqn. (5) for the rest of the paper.

The decay timescale (e-folding scale) is

$$t_{DEC} = \frac{2H}{r} \quad (6)$$

so that we expect the decay timescale to be longer in deeper lakes.

The quality factor Q (e.g. Tipler and Mosca, 2008, or any calculus-based physics textbook) is proportional to the ratio of these two timescales:

$$Q = \pi \frac{t_{DEC}}{t_{OSC}}$$

which in this case becomes

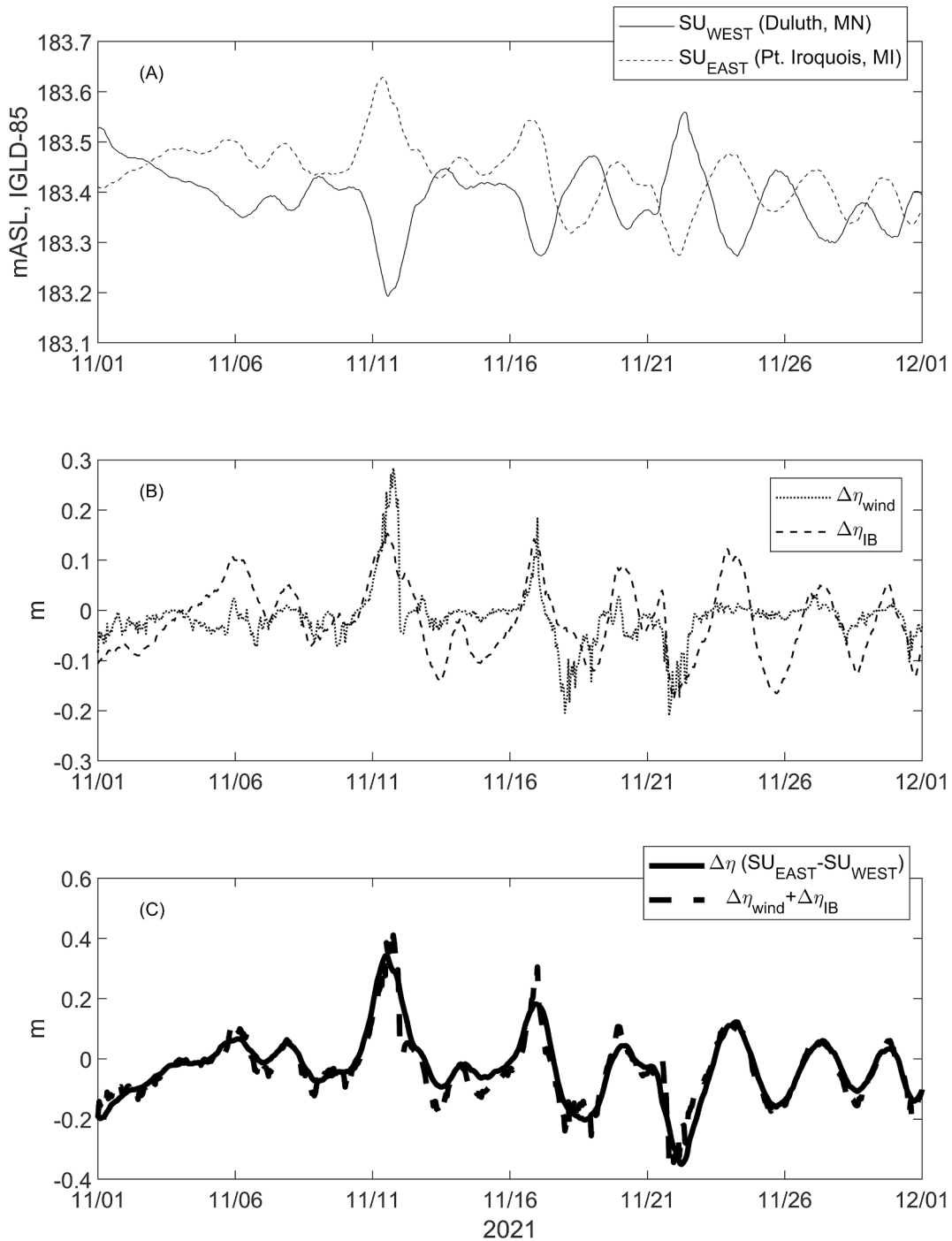


Fig. 2. An example of data from Lake Superior, November 2021. (A) Water levels at SU_{EAST} (Point Iroquois, MI) and SU_{WEST} (Duluth, MN), (B) predicted wind setup (eqn. (3)) and inverted barometer effect (eqn. (4)); (C) low-passed, observed lake level difference across the lake and the sum of the wind and inverted barometer effect.

$$Q = \frac{\pi \sqrt{gH^3}}{rL} \quad (7)$$

and is a measure of the duration of seiche behavior compared to its oscillation period. This suggests that deep lakes will have larger Q values, so that the lake will undergo several full oscillations before bottom drag damps out the signal. Shallower lakes, such as Erie, will have seiche behavior effectively damped out after just a few oscillations. Systems with values of $Q < 1/2$ are termed overdamped (Tipler and Mosca, 2008) and display no oscillatory behavior, in which case this analysis does not apply to these systems.

Data.

To evaluate these terms, we use atmospheric pressure data from Coastal Marine Automated Network (CMAN) (<http://ndbc.noaa.gov>; accessed August 2024) sites at either end of the lakes and a wind velocity measurement from a station close to the middle of each lake (Fig. 1; Table 1). Because we want to evaluate these terms for the whole year, surface meteorological buoys, which are recovered each year in the winter, cannot be used. The algorithm of Fairall et al. (1996) is used to estimate wind stress from wind speed. Water levels are available from the National Oceanic and Atmospheric Administration (NOAA) (<http://tidesandcurrents.noaa.gov>, accessed August 2024) and are taken from

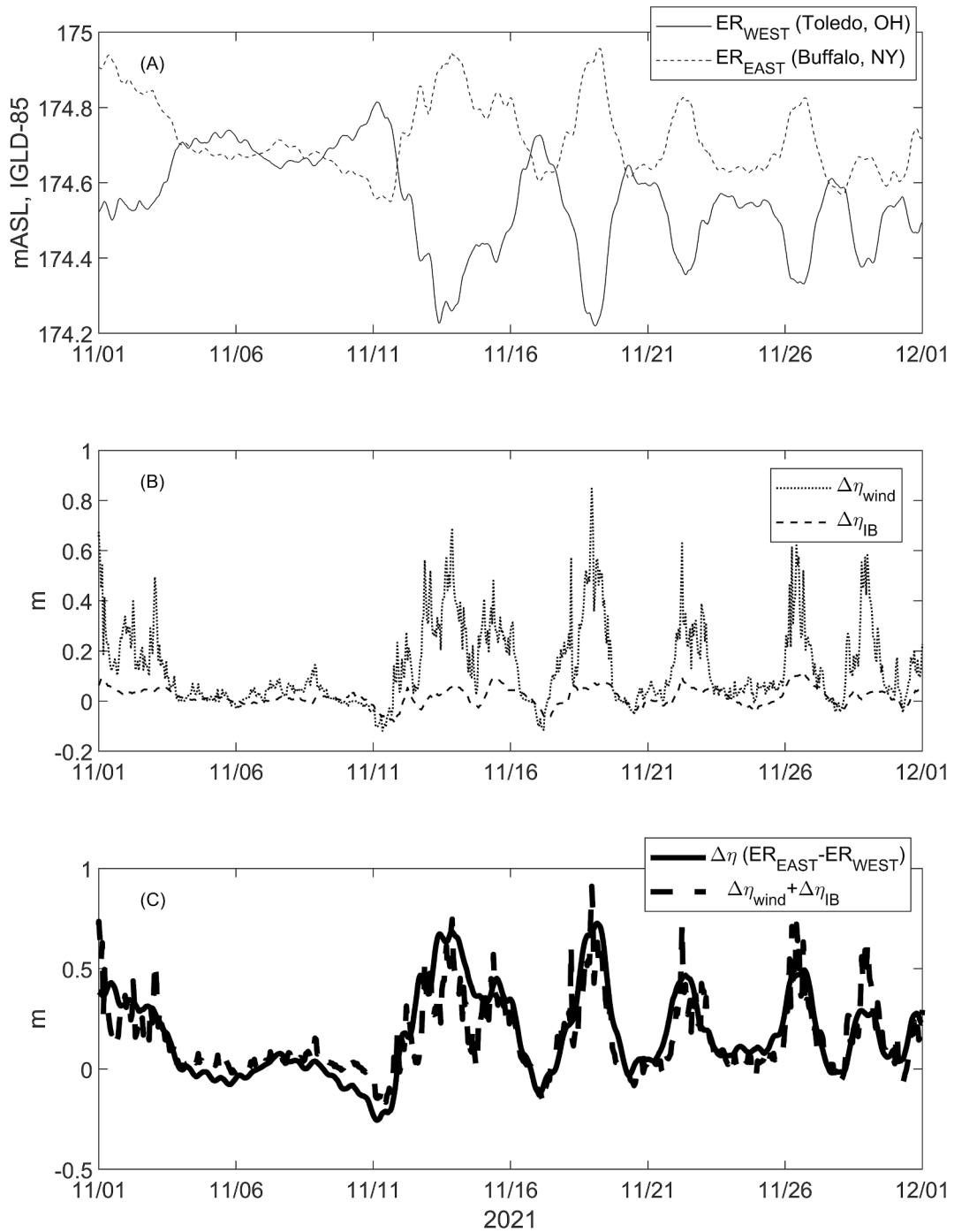


Fig. 3. An example of data from Lake Erie, November 2021. (A) Water levels at ER_{WEST} (Toledo, OH) and ER_{EAST} (Buffalo, NY), (B) predicted wind setup (3) and inverted barometer effect (4); (C) low-passed, observed lake level difference across the lake and the sum of the wind and inverted barometer effect.

the same locations as the atmospheric pressure data. All water levels are relative to the 1985 International Great Lakes Datum (IGLD-85), and have a temporal resolution of 6 min for all stations. For both the analysis of the forced and free response, the difference in lake levels between the ends of each lake along its major axis is used as a measure of surface displacement; this eliminates the effects of longer-term volumetric change such as seasonality and interannual variability:

$$\Delta\eta = \eta_2 - \eta_1 - \overline{(\eta_2 - \eta_1)} \quad (8)$$

The long-term average difference between the two ends of each lake is non-zero due to isostatic rebound since the establishment of the datum

(e.g. Clark et al., 1994) and is removed. For the analysis of the forced response, this difference is low pass filtered with a 24 h moving mean to eliminate the free oscillatory response. The straight-line distance between the stations where pressures are measured is used as the effective length of the lake, and the station locations are used to determine the orientation of the lake's primary axis (Fig. 1), onto which the wind stress is projected to get the along-lake component of the wind stress. The average depth of each lake is used for H (Table 1). All statistical analyses were done using data from the 10 year period 2014–2023.

Table 2

Properties of the forced response.

	$\langle \eta_{wind} \rangle (m)$	$\langle \eta_{IB} \rangle (m)$	$\langle \eta \rangle (m)$
Superior	0.036	0.057	0.083
Michigan	0.038	0.046	0.086
Huron	0.028	0.033	0.075
Erie	0.15	0.032	0.26
Ontario	0.017	0.021	0.031

3. Results

A month long, arbitrarily chosen sample of the water levels at opposite ends of lakes Superior and Erie (Fig. 2A and 3A) show a clear asymmetric response with the two sides of the lakes mirroring each other, is expected given the spatial structure of the primary mode. The estimated wind-induced setup (eqn. (3)) and inverted barometer setup (eqn. (4)) (Fig. 2B and 3B) show different properties between the two lakes, with the inverted barometer effect dominating in Superior and wind-induced setup dominating in Erie, although in the case of Superior

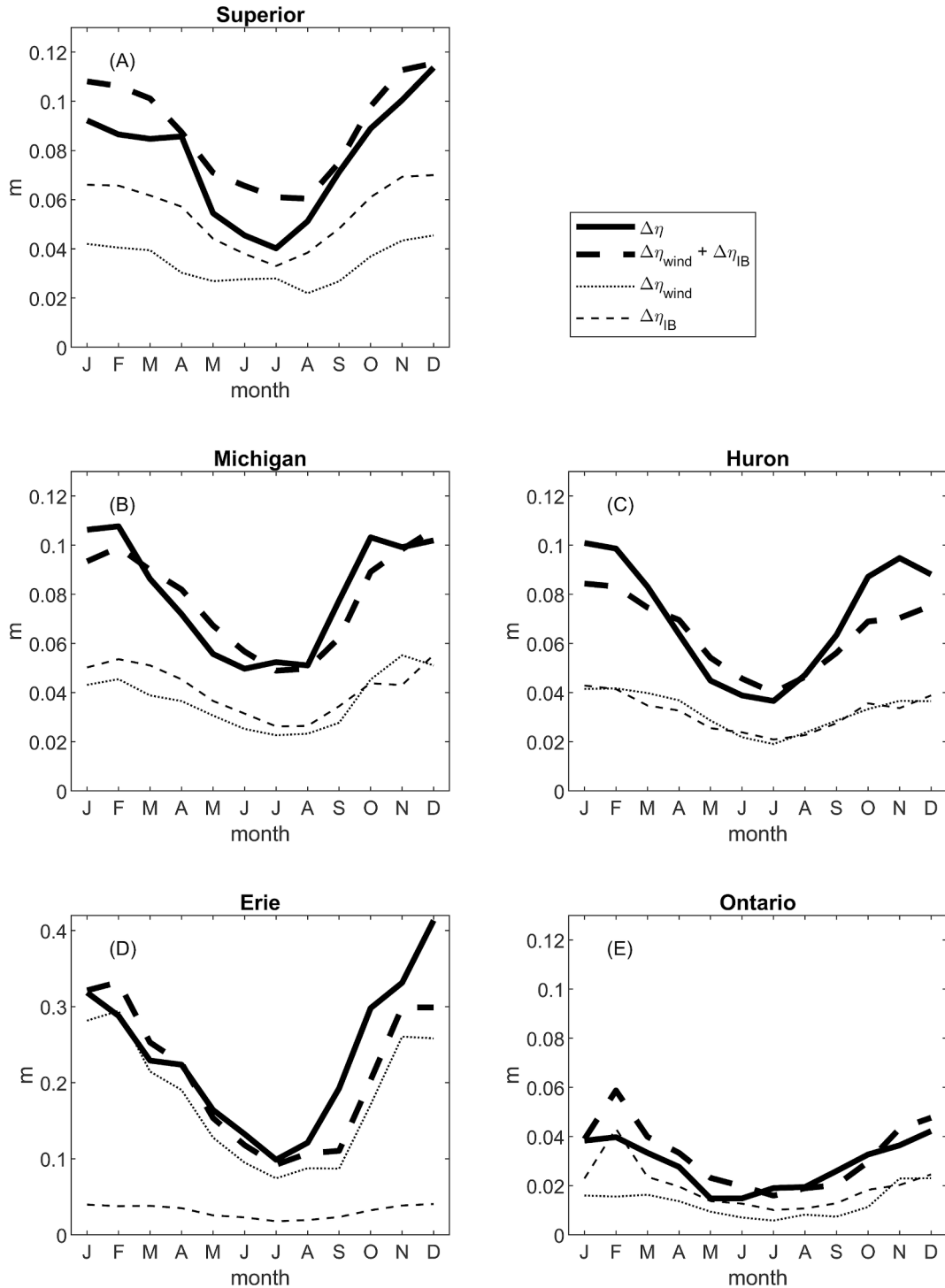


Fig. 4. Monthly variance of low-passed, observed lake level difference across lake, the two forcing terms, and the sum of the two forcing terms for (a) Lake Superior, (b) Michigan, (c) Huron, (d) Erie, and (e) Ontario. note that the vertical scale for lake Erie is different from the other four lakes.

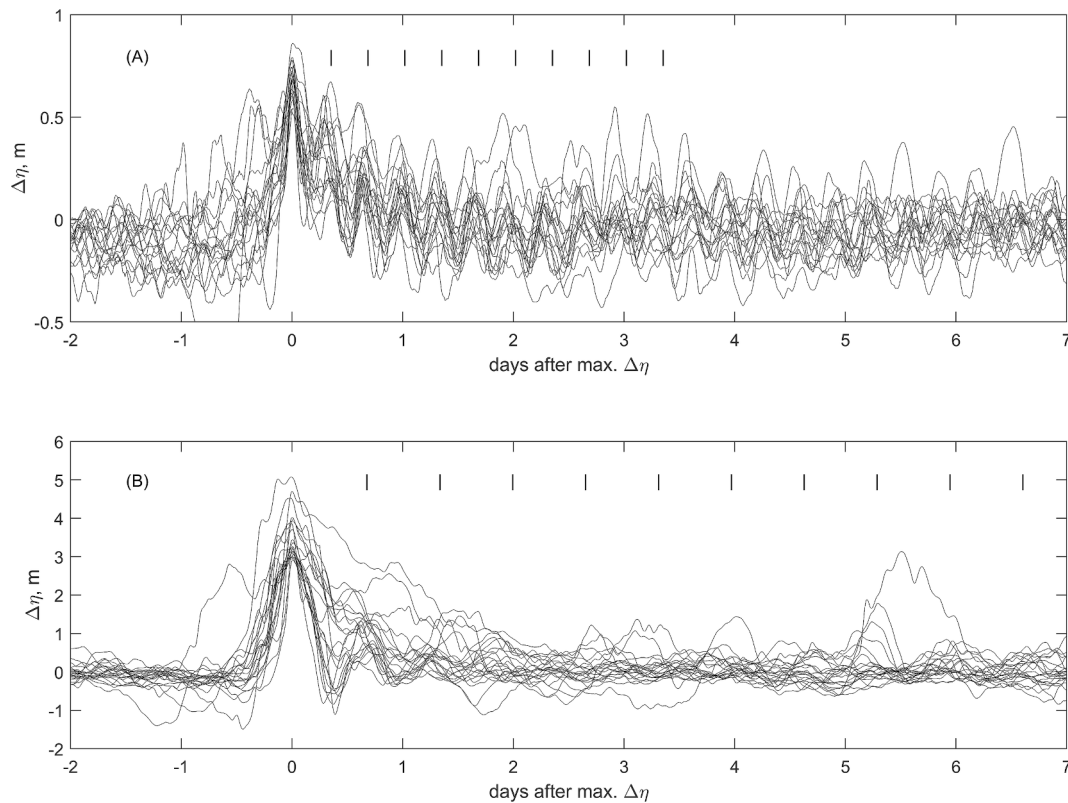


Fig. 5. Composite of events on (A) Lake Superior for which $\Delta\eta$ the lake level difference exceeded 0.6 m and (B) Lake Erie where $\Delta\eta$ exceeded 3 m. Tick marks on both panels mark multiples of the respective seiche period T_{OSC} .

the wind driven component can at times be significant. The low-passed difference in water level across the lakes along with the sum of the inverted barometer term and the wind stress term (Fig. 2C and 3C) show that the sum of the simple models does a remarkable job of capturing individual setup events, both in timing and magnitude. Overall, the response in Erie is much greater due to the much larger wind-driven response. Table 2 summarizes the contribution of each term for all five lakes.

To characterize seasonality, we can use monthly variance in pressure differences and in wind stress and to determine the contributions of the inverted barometer effect and wind-driven setup (Fig. 4; note that the y-axis is different for Lake Erie). There is a clear annual cycle across the great lakes, with larger response to both forcing terms in the winter. While the inverted barometer response is similar across the lakes (it is somewhat larger in Superior due to Superior's greater length), the importance of the wind-forced term varies significantly across the lakes. Specifically, wind forcing is dominant in Erie, due to its shallowness, but small relative to the inverted barometer effect in Superior. In Lakes Michigan, Huron, and Ontario, the two forcings are of similar importance year-round. In addition to the contributions from the two forcing terms, the variance of the observed water level difference across the lake is shown, along with the sum of the two forcing terms. The sum of the forcing terms and the observed water level difference compare remarkably well considering the simplicity of the models used. The exception to this is Lake Erie, especially in the winter. This is likely due to the fact that the wind stress term dominates, and there is uncertainty around the value of wind stress used; we are using wind speed from a single station to effectively determine stress over the entire lake. It has been noted (Schwab and Morton, 1984) that coastal sites may not adequately represent wind stress over the open waters of a lake, given the lateral scales of the lakes in question. The role of lake depth in determining the response is clear: setup in Erie is almost entirely wind driven; whereas, Superior's significantly smaller response is dominated

Table 3
Properties of the free response.

	t_{OSC} (h) (Eqn. (5))	Seiche period (h) (Measured)	Seiche period (h) (Modeled)	t_{DEC} (h) (Eqn. (6))	Q
Superior	8.2	7.90 ¹	7.84 ² 7.83 ³ 7.87 ⁷	163	62
Michigan	9.5	8.96 ¹		94	32
Huron	7.7	6.67 ¹		66	26
Erie	15.8	14.38 ⁴	14.86 ²	21	4.1
Ontario	6.0	5.06 ⁵	4.86 ⁶ 5.1 ³ 5.08 ⁵	96	51

¹ Mortimer et al. (1976).

² Platzman 1972.

³ Rao and Schwab 1976.

⁴ Platzman and Rao 1964.

⁵ Hamblin 1982.

⁶ Hamblin 1972.

⁷ Kelly et al 2024.

by the inverted barometer effect. For the other three lakes, the two forcings are similar in magnitude.

A free oscillatory seiche response typically follows forcing events, as the lake surface returns to an equilibrium state and overshoots. The character of the seiche differs between the lakes. Water level difference for 20 large setup events (>0.6 m for Superior; >3 m for Erie) is plotted in terms of the amount of time elapsed since maximum water level (Fig. 5). In Superior, multiple full cycles of the seiche are typically observed, whereas one full oscillation at best is observed for Erie. The difference between these responses is a function of the relative values of the time scale of oscillation (eqn. (5)) and the decay time scale (eqn. (6)). In Superior, the decay time scale is about 20 times the oscillation

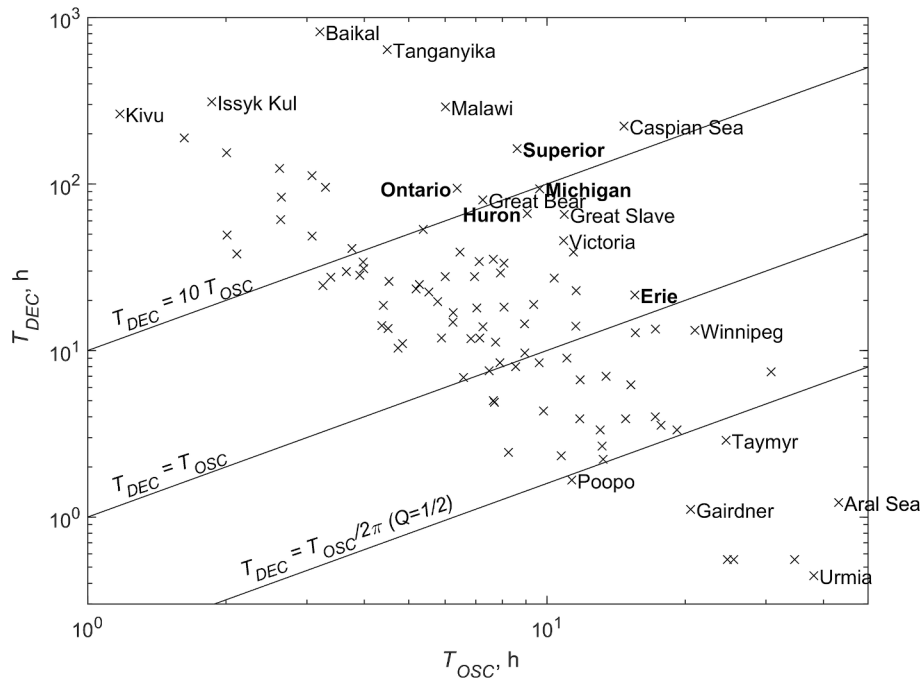


Fig. 6. Decay and oscillation time scales for 100 lakes from a list of largest lakes by surface area. Names are placed on the top 15 lakes by surface area as well as some outliers.

time scale (Table 3), whereas on Erie the decay time scale is on the same order as the time for one full oscillation. This suggests that the energy associated with the seiche in Lake Erie is quickly dissipated by bottom drag; in the other four lakes the decay timescale is significantly longer than the oscillation timescale (Table 3), leading to sustained seiching events. In addition, the estimated seiche period for Erie (15.8 h) is close to the local inertial period (approximately 18 h), so rotational effects likely play a role, but are ignored here, as incorporating rotational effects necessarily requires modeling the lakes in two dimensions and moves us away from simple analytical solutions.

4. Discussion and Conclusions

The simple model described here does a reasonable job at predicting the primary (mode 1) seiche period, although (unsurprisingly) does not yield estimates as accurate as from more sophisticated modeling approaches (Table 3). Estimates of observed seiche periods exist for all five lakes, and modeled estimates of the seiche period are available for Superior, Erie, and Ontario. In general, the seiche period is predicted to within a few percent by the well-resolved models, whereas the simple scaling presented here tends to consistently overestimate the period by 10–15 %. There do not appear to be systematic estimates of seiche decay rates across the Great Lakes, although Kelly et al (2024) estimate a decay time scale of 4.5 days, similar to what the simple model predicts for Superior. The more complex models also are capable of determining spatial structure of the primary (and often higher order) seiche, demonstrating the rotational nature of the response, whereas the simple model presented here reveals nothing about the spatial structure of the seiche other than that the two ends of a lake should be 180° out of phase with each other.

While characterizing wind-driven setup is extensively covered in the Great Lakes literature, the inverted barometer effect, while small for Erie, dominates Superior's response. In addition, while Erie may have the largest magnitude response to wind events, the resulting relaxation back to equilibrium does not have a very seiche-like behavior, due to the importance of bottom drag in a shallow lake. The depth-dependence of the wind driven response (and the lack of depth dependence for the inverted barometer response) is a function of the nature of the two forces

in question. Wind stress is a contact force (per unit area); it is imposed entirely at the surface, and as a lake gets deeper, this force is distributed over a greater volume of water, diminishing its impact in a deep lake. However, pressure gradients due to a surface slope or an atmospheric pressure gradient generate a body force; hydrostatics distributes the force due to the pressure gradient throughout the water column, so that regardless of how deep the water is the impact is the same. Bottom drag is also a contact force, making it more important in shallow lakes than in deep ones.

While the magnitude of wind-driven storm surges is by far the largest on Erie by virtue of its shallowness, it is also the only one of the lakes for which Q is relatively small and the seiching signal is damped out quickly. An analysis of lakes around the world with large surface areas (Fig. 6) was conducted using lake length and mean depth. Lake morphometry data was taken from the HydroLakes database (Messenger et al., 2016). In many cases where lake coastal morphometry is complex, the concept of “length” is ambiguous (Rueda and Schladow, 2002); we simply use the two most distant points on the coastline to estimate length, remembering that the goal here is not to establish quantitative values as much as it is to demonstrate that the character of the seiche response can vary across lakes worldwide. Erie is the one Laurentian Great Lake for which the decay timescale is close to the oscillatory period, a quality it shares with many lakes worldwide. A small number of large lakes worldwide fall below the $Q = 1/2$ line, which suggests that they should be overdamped by bottom drag and hence should not display any seiching behavior whatsoever (note that for these lakes, the T_{osc} time scale is physically meaningless as overdamped systems do not have any oscillatory behavior). At the other end of the spectrum an array of very deep lakes have damping times considerably in excess of their seiche period, meaning that they should display persistent seiching behavior, although the magnitude of the seiche response in these deep lakes is likely small. While the character of the free seiche response can be compared between lakes using just morphometric information, comparing the forced response (from either wind forcing or pressure gradients) between lakes worldwide is more difficult, since the meteorology over lakes distributed globally is likely to vary. Interestingly, because the magnitude of the wind setup is proportional to L/H (eqn. (3)), whereas the Q value is proportional to $H^{3/2}/L$ (eqn. (7)), it is

unlikely that there are any lakes that have both a large wind driven response followed by a sustained seiche response. Instead, lakes can have a large wind driven surge followed by a quickly dampened seiche (such as Erie), or a weak wind-driven response followed by a sustained seiche response (such as Superior).

Credit authorship contribution statement

Jay Austin: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Data availability

Meteorological data (barometric pressure and wind velocity) data are available through the National Data Buoy Center (ndbc.noaa.gov). Water level data are available through the National Oceanic and Atmospheric Administration's Center for Operational Oceanographic Products and Services (tideaandcurrents.noaa.gov). Code to produce figures 2–6 will be posted to the Data Repository for the University of Minnesota (<https://hdl.handle.net/11299/166578>). A digital object identifier (doi) for this submission is pending.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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